

4. (a) Calcium and oxygen react forming calcium oxide.

- (i) Explain, using dot and cross diagrams, how bonding takes place during the formation of calcium oxide. [2]

(ii)

Substance	Melting point (°C)
calcium oxide	2613
sodium chloride	801

Explain why both of these compounds have high melting points and why the melting point of calcium oxide is higher. [2]

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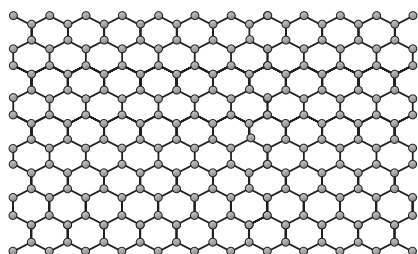
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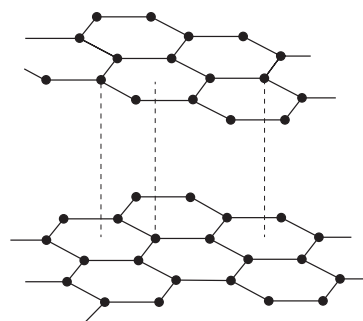
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(b) Graphene and graphite are different forms of carbon.



graphene



graphite

(i) Explain how the bonding present in both forms makes them good conductors of electricity. [2]

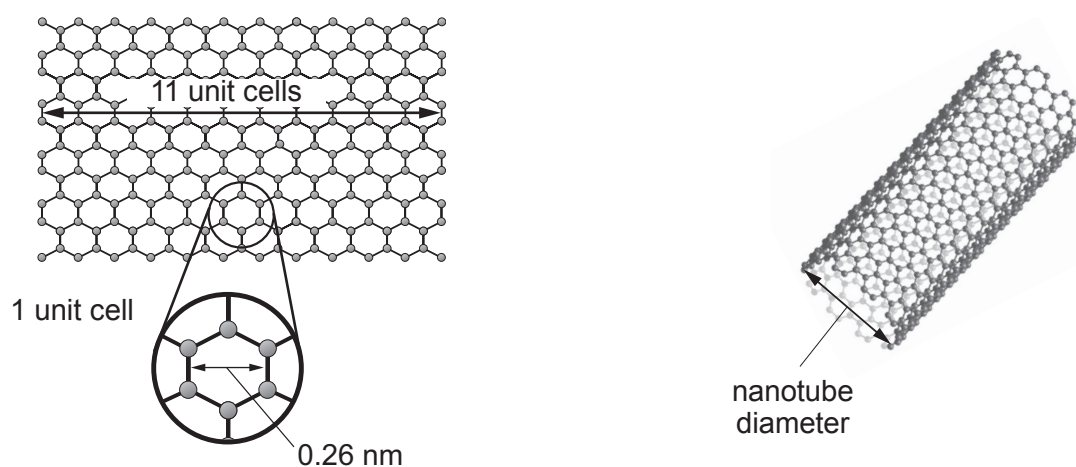
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(ii) Carbon nanotubes can be formed by rolling up graphene sheets.



A sheet of graphene 11 unit cells in width was rolled up to form a nanotube. Use the information below to calculate the diameter, in metres, of the nanotube formed. Write your answer in **standard form**. [3]

$$\text{circumference} = \pi \times \text{diameter}$$

$$\pi = 3.14$$

$$1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

Diameter = m

